

FreeBSD SOC Setup: Suricata + Cowrie Honeypot in a Jail

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1. Introduction

This guide details building a lightweight mini-SOC on FreeBSD 14.3, combining:

- **Suricata** (IDS/IPS) for network detection
- **Cowrie** (SSH/Telnet honeypot) to capture attacks
- **Jail-based isolation** for security

2. Prerequisites

- FreeBSD 14.3 installed
- Root access or a user with sudo
- Active internet access to download packages
- Python ≥ 3.11

3. System Update & Dependency Installation

3.1 Update FreeBSD

```
sudo pkg update
sudo pkg upgrade
```

3.2 Install Essential Tools

```
sudo pkg install gcc gmake libffi pkgconf rust git python311 py311-venv bash vim htop
```

4. Installing & Configuring Suricata

4.1 Install Suricata

```
sudo pkg install suricata
sudo sysrc suricata_enable="YES"
sudo service suricata start
```

4.2 Configure Suricata to Listen on wlan0

Edit `/usr/local/etc/suricata/suricata.yaml` :

```
default-rule-path: /var/lib/suricata/rules/rules
rule-files:
- emerging.rules
af-packet:
- interface: wlan0
```

4.3 Download Rules (e.g. Emerging Threats)

```
sudo mkdir -p /var/lib/suricata/rules/rules
cd /var/lib/suricata/rules/rules
sudo fetch https://rules.emergingthreats.net/open/suricata/emerging.rules.tar.gz
sudo tar -xvzf emerging.rules.tar.gz
sudo rm emerging.rules.tar.gz
sudo service suricata restart
```

5. Installing Cowrie (SSH/Telnet Honeypot)

5.1 Clone the Cowrie Repository

```
cd /usr/local
sudo git clone https://github.com/cowrie/cowrie.git
cd cowrie
```

5.2 Create a Python Virtual Environment

```
sudo python3.11 -m venv cowrie-env
sudo chown -R $(whoami) cowrie-env
. cowrie-env/bin/activate
```

5.3 Install Python Dependencies

```
pip install --upgrade pip setuptools wheel
pip install -r requirements.txt
```

6. Configuring Cowrie

6.1 Copy and Edit `cowrie.cfg`

```
cp etc/cowrie.cfg.dist etc/cowrie.cfg
```

Edit `etc/cowrie.cfg` :

```
[honeypot]
hostname = svr04
listen_port = 2222
telnet_port = 2223
log_path = var/log/cowrie
download_path = var/lib/cowrie/downloads
```

6.2 Create the Log Directories

```
mkdir -p var/log/cowrie var/lib/cowrie/downloads
chmod -R 755 var/log/cowrie var/lib/cowrie
```

6.3 Start Cowrie

```
bin/cowrie start
```

7. Isolation with Jails (FreeBSD)

7.1 Create a Jail for Cowrie

```
sudo pkg install ezjail
sudo ezjail-admin create cowrie-jail 'lo1|127.0.0.2'
sudo ezjail-admin start cowrie-jail
sudo ezjail-admin console cowrie-jail
```

7.2 Prepare the Jail for Cowrie

Inside the jail, repeat steps 3 and 4 to install Cowrie.

8. Jail Configuration for Cowrie

8.1 Create the Jail Directory Structure

```
sudo mkdir -p /opt/cowrie-jail/bin
sudo mkdir -p /opt/cowrie-jail/lib
sudo mkdir -p /opt/cowrie-jail/libexec
sudo mkdir -p /opt/cowrie-jail/etc
sudo mkdir -p /opt/cowrie-jail/usr
sudo mkdir -p /opt/cowrie-jail/var
sudo mkdir -p /opt/cowrie-jail/tmp
sudo mkdir -p /opt/cowrie-jail/dev
sudo chmod 1777 /opt/cowrie-jail/tmp
```

8.2 Copy Binaries and Libraries

```
sudo cp /bin/sh /opt/cowrie-jail/bin/
sudo cp /bin/ls /opt/cowrie-jail/bin/
sudo cp /bin/cat /opt/cowrie-jail/bin/
sudo cp /bin/echo /opt/cowrie-jail/bin/
sudo cp /bin/pwd /opt/cowrie-jail/bin/
sudo cp /bin/date /opt/cowrie-jail/bin/
sudo cp /bin/mkdir /opt/cowrie-jail/bin/
sudo cp /bin/rm /opt/cowrie-jail/bin/
sudo cp /usr/bin/whoami /opt/cowrie-jail/bin/
sudo cp /usr/bin/id /opt/cowrie-jail/bin/
sudo cp /usr/bin/env /opt/cowrie-jail/bin/
```

8.3 Copy Libraries

```
sudo cp /lib/libc.so.7 /opt/cowrie-jail/lib/
sudo cp /lib/libutil.so.9 /opt/cowrie-jail/lib/
sudo cp /lib/libedit.so.8 /opt/cowrie-jail/lib/
sudo cp /lib/libtinfow.so.9 /opt/cowrie-jail/lib/
sudo cp /libexec/ld-elf.so.1 /opt/cowrie-jail/libexec/
```

8.4 Create Device Nodes in `/dev`

```
sudo mknod /opt/cowrie-jail/dev/null c 0 0
sudo mknod /opt/cowrie-jail/dev/zero c 0 12
sudo mknod /opt/cowrie-jail/dev/tty c 0 3
sudo chmod 666 /opt/cowrie-jail/dev/null
sudo chmod 666 /opt/cowrie-jail/dev/zero
sudo chmod 666 /opt/cowrie-jail/dev/tty
```

8.5 Add Configuration Files

```
sudo sh -c 'echo "root:x:0:0:root:/root:/bin/sh" > /opt/cowrie-jail/etc/passwd'
sudo sh -c 'echo "root:*:0:0:root:/root:/bin/sh" > /opt/cowrie-jail/etc/master.passwd'
sudo sh -c 'echo "wheel:*:0:root" > /opt/cowrie-jail/etc/group'
```

8.6 Copy Cowrie into the Jail

```
sudo cp -r /usr/local/cowrie /opt/cowrie-jail/usr/
```

8.7 Test the Jail

```
sudo chroot /opt/cowrie-jail /bin/sh
```

9. Testing & Validation

9.1 Confirm Cowrie Is Listening on Port 2222

```
sockstat -4 -l | grep 2222
```

9.2 Test an SSH Connection

```
ssh root@<SERVER_IP> -p 2222
```

9.3 Check the Logs

```
tail -f /usr/local/cowrie/var/log/cowrie/cowrie.json
```

10. Final Directory Layout

```
/usr/local/  
├── cowrie/  
│   ├── bin/  
│   ├── cowrie-env/  
│   ├── var/  
│   │   ├── log/cowrie/  
│   │   └── lib/cowrie/  
│   └── etc/cowrie.cfg  
└── suricata/
```

11. Important Notes

- **Suricata:** listens on `wlan0` and uses the Emerging Threats ruleset.
- **Cowrie:** listens on ports 2222 (SSH) and 2223 (Telnet).
- **Logs:** centralized in `/var/log/suricata/` and `/usr/local/cowrie/var/log/cowrie/`.
- **Security:** for a production-grade lab, use a dedicated jail or VM.

12. Conclusion

Setting up this lab took about 8 hours. It was genuinely interesting, though not a trivial exercise — the commands above are the ones that actually got the lab working. If something unexpected happens on your end, check the official documentation or ask an LLM for help; that's what got me unstuck more than once.

To avoid putting all your eggs in one basket — which matters a lot in security if a machine gets compromised or goes down — the SIEM (Splunk) and Samba components are covered separately, on a different machine, in another guide.